Chapter 02

Managing Hardware, Storage, and Filesystems

Episode 2.01

Episode title: **Device & Kernel Modules**

Objective: 1.2

Device & Kernel Modules Overview

Linux abstracts hardware

Device files

Kernel modules

Linux Device Files

Hardware represented as files

Located in `/dev`

Block vs character devices

Linux Device Visibility

Linux represents hardware through virtual filesystems

`/dev` – Device nodes (block and character devices)

`/proc` – Runtime kernel and process information

`/sys` – Hardware and driver information

Exam focus: knowing when to check each

Udev and Dynamic Devices

udev manages dynamic device creation

Enables persistent device naming

Supports hot-plug devices (USB, disks, NICs)

udev rules stored in `/etc/udev/rules.d/`

Kernel Modules

Extend kernel functionality

Load without rebooting

- depmod
- insmod
- lsmod
- modinfo
- modprob
- rmmod

Device Management

Device management in Linux

Kernel and hardware visibility

- dmesg
- dmidecode
- ipmitool
- lm sensors
- lscpu
- lshw
- lsmem
- lspci
- lsusb

initrd Management

initrd / initramfs purpose

Early boot hardware support

dracut

mkinitrd

When to rebuild initrd

Custom Hardware & Specialized Systems

Custom hardware environments

Embedded systems

GPU-accelerated workloads

Graphics Processing Unit (GPU) use cases

nvtop

Episode 2.03

Episode title: **Storage Concepts & Partitions**

Objective: 1.3

Storage Fundamentals

Block devices

Disk naming

Storage layouts

Partitions and Partition Tables

Purpose of partitions

MBR vs GPT

Partition alignment

Partitioning Tools – Exam Coverage

- `lsblk` – View block devices and layout
- `blkid` – Identify filesystem type and UUID
- `sfdsik / gdisk` – MBR and GPT partitioning
- `growpart` – Expand partitions on cloud disks

Partitioning with Parted (DEMO)

lsblk

parted

Non-destructive changes

Episode 2.04

Episode title: **LVM Essentials**

Objective: 1.3

LVM Architecture

Physical volumes

Volume groups

Logical volumes

Why Use LVM?

Flexible sizing

Easier expansion

Enterprise standard

LVM Display & Inspection Commands

High-level views:

- pvs, vgs, lvs

Detailed views:

- pvdisplay
- vgdisplay
- lvdisplay

Safe, read-only commands

LVM Management Commands

Logical volumes:

lvchange, lvremove, lvresize

Volume groups:

vgextend, vgremove

vgexport, vgimport

Physical volumes:

pvmove, pvresize, pvremove

Discovery:

vgscan, pvscan

Creating and Resizing LVM Volumes (DEMO)

pvcreate

vgcreate

lvcreate

lvextend

resize2fs / xfs_growfs

Episode 2.05

Episode title: **RAID & Redundancy**

Objective: 1.3

Why RAID?

Performance

Fault tolerance

Availability

Common RAID Levels

RAID 0

RAID 1

RAID 5

RAID 10

RAID Status & Monitoring

Check RAID status with `/proc/mdstat`

Identify degraded arrays

Used alongside `mdadm`

Common troubleshooting scenario

Software RAID with mdadm (DEMO)

mdadm

Create array

Monitor and manage RAID

Episode 2.06

Episode title: **Filesystem Types & Tools**

Objective: 1.3

Common Linux Filesystems

ext4

XFS

Btrfs

Filesystem Creation and Management

Formatting

Metadata

Maintenance

Inodes and Filesystem Limits

Inodes store file metadata

Filesystem can run out of inodes

Disk space may still be available

Check inode usage with `df -1`

Filesystem Utilities – Exam Tools

Space usage:

- df, du

Filesystem checks:

- fsck, xfs_repair

Resizing:

- resize2fs, xfs_growfs

Performance testing:

- fio

Episode 2.07

Episode title: **Filesystem Types & Tools Demo**

Objective: 1.3

Filesystem Utilities (DEMO)

mkfs

fsck

tune2fs

Episode 2.08

Episode title: **Mounting & Network Storage**

Objective: 1.3

Filesystem Mounting Concepts

Mount points

Temporary vs permanent mounts

Directory tree integration

Persistent Mounts and /etc/fstab

/etc/fstab

Mount options

Automounting basics

Mount Information Sources

`/etc/mtab` – Legacy mount tracking

`/proc/mounts` – Kernel-reported mounts

`/proc/mounts` is authoritative

Used for troubleshooting mount issues

Mount Options – Exam Favorites

Performance:

noatime, nodiratime

Security:

nodev, nosuid, noexec

Reliability:

nofail

Access control:

ro, rw, remount

Automounting with autofs

Automatically mounts filesystems on access

Reduces boot-time dependencies

Common for network storage

Managed by the `autofs` service

Mounting Local Filesystems (DEMO)

mount

umount

Verify mounts

Mounting Network Storage (NFS/SMB) DEMO

mount.nfs

mount.cifs

Verify access

Network Storage Concepts

NFS

SMB/CIFS

Use cases